THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is correct for compounds? 1

(a) constituents cannot be separated by simple physical methods
(b) they have no fixed composition
(c) they are elements
(d) they are mixtures

2. Assertion (A): Mercury is used in thermometers. Reason (R): Mercury is a metal that is liquid at room temperature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. A non-metal oxide dissolved in water generally gives a solution that turns: 1

(a) turmeric red **(b)** red litmus blue **(c)** red litmus red **(d)** blue litmus red

4. A mixture of chalk powder and water is shaken and left. The chalk settles. This shows the mixture is: 1

(a) a solution **(b)** an element
(c) a compound **(d)** a suspension (heterogeneous)

5. Assertion (A): Diamond and graphite have different properties although they are both made of carbon. Reason (R): Their atoms are arranged differently. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Which of the following will be a good thermal insulator? 1

(a) iron **(b)** aluminium **(c)** copper **(d)** wood

7. Which of the following is not a physical method of separation? 1

(a) magnetic separation **(b)** evaporation
(c) filtration **(d)** electrolysis of water

8. Assertion (A): Aluminium foil can be made from aluminium metal. Reason (R): Aluminium is malleable. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Water is a mixture of hydrogen and oxygen. 1
Reason (R): Water has a fixed composition and properties different from those of hydrogen and oxygen.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Sulphur is brittle and has no lustre. 1
Reason (R): Non-metals are generally brittle and not shiny.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. An element X reacts with oxygen to form an oxide that is acidic. Is X a metal or non-metal? Give an example. 2

12. Write the number and type of atoms in one molecule of glucose, $C_6H_{12}O_6$. 2

13. State two limitations of classifying elements only as metals and non-metals. 2

14. State why an iron nail dipped in copper sulphate becomes brown and the solution fades. Name the type of change. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a brass key does not rust but an iron nail does. 3

16. Explain why brass and bronze are not elements and not compounds. 3

17. Explain why the electrical wires are usually made of copper while the outer cover is plastic. 3

OR

List the steps to separate a mixture of iron filings, ammonium chloride and sand with the property used in each. 3

18. Explain why mercury, bromine and gallium are special among elements. 3

OR

Why do we say that air, sea water and soil are mixtures but not pure substances? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In an activity, iron filings and sulphur powder are mixed and a magnet is passed over the mixture; the iron is pulled out. In the second part, the mixture is heated strongly in a test tube; a black solid, iron sulphide, is formed, which a magnet does not attract. 5
(i) Is the first sample a mixture or a compound?
(ii) Name the compound formed on heating.
(iii) Does the magnet attract the compound?
(iv) Give two differences between a mixture and a compound shown by this activity.

20. Compare physical changes with chemical changes using examples of mixtures and compounds. Include dissolving, heating iron-sulphur, rusting and melting wax. 5

OR

A sample of ammonium chloride and sand is given. Describe the separation and explain the property used. 5

END OF QUESTION PAPER — Total: 40 Marks